

St. John, USVI - Post-Hurricane Irma Community Survey

Buildings & Civic Property · 2026-09-30

OVERALL CONDITION	17	15	2	0
CRITICAL	Findings	P1 Immediate	P2 Planned	P3 Monitor

Source footage	civic_hurricane_irma_stjohn.webm
Footage length	40.4 s
Frames analysed	13 (one every 3.0 s)
Raw observations	21 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	6.1 s

How urgent is each priority

P1	Immediate (within 24-72 h)
P2	Planned (within 30 days)
P3	Scheduled / monitor (next maintenance cycle)

1. Executive summary

US Navy helicopter footage filmed on 13 September 2017, one week after Hurricane Irma, over Coral Bay and Cruz Bay on St. John, US Virgin Islands. Damage is widespread rather than isolated: across the Coral Bay hillsides many homes have lost roofs or upper floors (F-01, F-07, F-05), some are largely destroyed (F-02, F-14), and a large masonry building near the ballfield is reduced to roofless walls (F-04). Debris covers the slopes between homes (F-12) and dozens of boats are wrecked along the harbour shore (F-03). In Cruz Bay most larger buildings kept their roofs, but several roofs are stripped to the rafters (F-08, F-09) or gone (F-10), and a cluster of small structures has collapsed (F-06). The public water storage tank shows no obvious shell damage from the air but needs inspection before its water is trusted (F-16). Nearly every finding is P1 because this is an active disaster zone; life safety comes first.

Key risks

- People sheltering in or entering damaged buildings that could collapse (F-02, F-04, F-06).
- Open roofs letting rain destroy what is left of homes and create mould (F-07, F-08, F-09, F-10).
- Loose debris becoming projectiles in the next storm and blocking roads and drainage (F-12).
- Fuel and oil leaking from wrecked boats into the harbour (F-03, F-15).
- Drinking water contamination if the tank roof or vents were damaged (F-16).

Findings by category



2. Action plan

Immediate (24-72 hours)

- Run house-by-house rapid safety evaluations and placard unsafe buildings; move residents of unsafe homes to shelter (F-05).
- Fence off the ruined masonry building and destroyed homes (F-04, F-02, F-14).
- Tarp open roofs on occupied homes before the next rain (F-07, F-08, F-09, F-10).
- Inspect the water tank roof, hatches and vents and test the water before use (F-16).
- Boom off leaking vessels in Coral Bay harbour (F-03).

Next 30 days

- Clear debris from roads, drains and around homes, separating hazardous material (F-12, F-17).
- Salvage wrecked and stranded boats and survey the harbour for sunken hazards (F-03, F-15).
- Engineer's decision on repair versus demolition for each destroyed building.

Next 90 days

- Re-roof damaged homes with hurricane-rated fixings; prioritise occupied homes and community buildings.
- Re-fly the same routes to track recovery and confirm buildings are weather-tight before the next storm season.

3. Findings register

All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	00:00.0	House with roof and upper structure torn open	roof damage	Critical (4.0)	P1	Keep out / make safe
F-02	00:03.0	House largely destroyed	structural damage	Critical (4.0)	P1	Keep out / make safe
F-03	00:09.0	Boats wrecked and piled up along the harbour shore	debris and loose material	Critical (4.0)	P1	Keep out / make safe
F-04	00:15.0	Large building reduced to roofless walls	structural damage	Critical (4.0)	P1	Keep out / make safe
F-05	00:21.0	Widespread damage across hillside neighbourhood	roof damage	Critical (4.0)	P1	Keep out / make safe
F-06	00:30.0	Cluster of small structures collapsed	structural damage	Critical (4.0)	P1	Keep out / make safe
F-07	00:03.0	Roof blown off, top-floor framing exposed	roof damage	Critical (3.9)	P1	Keep out / make safe
F-08	00:30.0	Roof stripped to exposed rafters	roof damage	Critical (3.9)	P1	Keep out / make safe
F-09	00:30.0	Roof stripped to exposed rafters	roof damage	Critical (3.9)	P1	Keep out / make safe
F-10	00:33.0	Building roof gone, interior exposed	roof damage	Critical (3.9)	P1	Keep out / make safe
F-11	00:33.0	Informal structures with damaged roofs	roof damage	Critical (3.8)	P1	Keep out / make safe
F-12	00:00.0	Hurricane debris field across hillside homes	debris and loose material	Critical (3.7)	P1	Clean
F-13	00:33.0	House damaged with debris around the lower floor [verify on site]	debris and loose material	Critical (3.7)	P1	Clean
F-14	00:15.0	Homes destroyed beside the main road	debris and loose material	Critical (3.6)	P1	Keep out / make safe
F-15	00:24.0	Boat stranded on the beach	debris and loose material	Major (3.0)	P1	Keep out / make safe
F-16	00:36.0	Public water storage tank needs post-storm inspection	utilities damage	Major (2.8)	P2	Keep out / make safe
F-17	00:24.0	Debris washed up along the beach and waterfront	debris and loose material	Moderate (2.1)	P2	Clean

4. Detailed findings

F-01 House with roof and upper structure torn open

P1 - Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:00.0 (seen in 1 frame(s), from 00:00.0)
Component	multi-storey house, upper right
Extent	1 house, top floor
Category	roof damage (100% confidence)
Severity	Critical, score 3.95 / 4
Safety / environmental hazard	86% probability
Likelihood it is a real defect	94%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A multi-storey house has lost part of its roof and upper walls, leaving the upper floor open to the weather.

Visual evidence: Missing roof sections and exposed interior on the top floor, debris spilling down the slope below.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-02 House largely destroyed

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:03.0 (seen in 1 frame(s), from 00:03.0)
Component	house, upper right
Extent	1 house
Category	structural damage (63% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	90% probability
Likelihood it is a real defect	87%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Structural engineer + building official; demolition or shoring contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A house at the top right has lost its roof and most of its upper structure.
Visual evidence: Only parts of walls and a platform remain; debris around it.

How to fix it

For this finding: Placard the building unsafe and keep people out until a structural engineer assesses it; plan demolition if beyond repair.

Standard procedure

- 1 Cordon off a fall zone around the structure (at least its height) and post it as unsafe.
- 2 Get a structural engineer's rapid safety evaluation (inspected / restricted use / unsafe placard).
- 3 Shore or brace if the structure is to be saved; otherwise plan a controlled demolition.
- 4 Do not let anyone climb or enter until the engineer clears it.

References: ATC-45 (safety evaluation after windstorms and floods); FEMA P-2055 (post-disaster building safety evaluation); IEBC

F-03 Boats wrecked and piled up along the harbour shoreP1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:09.0 (seen in 2 frame(s), from 00:09.0)
Component	Coral Bay harbour waterfront and boatyard
Extent	Along ~300 m of shoreline
Category	debris and loose material (96% confidence)
Severity	Critical, score 3.99 / 4
Safety / environmental hazard	93% probability
Likelihood it is a real defect	89%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Dozens of boats have been driven ashore, sunk or piled together along the harbour edge and in the boatyard. Wrecked boats leak fuel and oil and block the waterfront.

Visual evidence: Tight clusters of white hulls on the shore and in shallow water, some capsized, next to the waterfront buildings.

How to fix it

For this finding: Boom off leaking vessels, survey the harbour for sunken hazards, and organise vessel salvage before reopening the waterfront.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
 - 2 Clear debris from access routes, drains and around buildings.
 - 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.
- References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

F-04 Large building reduced to roofless walls

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:15.0 (seen in 2 frame(s), from 00:12.0)
Component	large masonry building near the ballfield
Extent	1 large building
Category	structural damage (53% confidence)
Severity	Critical, score 3.99 / 4
Safety / environmental hazard	87% probability
Likelihood it is a real defect	96%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Structural engineer + building official; demolition or shoring contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Closer view of the same ruined building: roof gone, walls partly collapsed.
Visual evidence: Standing masonry walls with no roof; debris field beside it.

How to fix it

For this finding: Fence off a fall zone, then structural evaluation.

Standard procedure

- 1 Cordon off a fall zone around the structure (at least its height) and post it as unsafe.
- 2 Get a structural engineer's rapid safety evaluation (inspected / restricted use / unsafe placard).
- 3 Shore or brace if the structure is to be saved; otherwise plan a controlled demolition.
- 4 Do not let anyone climb or enter until the engineer clears it.

References: ATC-45 (safety evaluation after windstorms and floods); FEMA P-2055 (post-disaster building safety evaluation); IEBC

F-05 Widespread damage across hillside neighbourhood

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:21.0 (seen in 1 frame(s), from 00:21.0)
Component	scattered homes across the hillside
Extent	Dozens of homes over roughly 1 km2
Category	roof damage (95% confidence)
Severity	Critical, score 3.98 / 4
Safety / environmental hazard	86% probability
Likelihood it is a real defect	85%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Across this neighbourhood many homes show missing roofs and debris fields around them; the damage is spread over the whole area, not just a few houses.

Visual evidence: Many small buildings with light-coloured debris around them across the brown slope.

How to fix it

For this finding: Do a house-by-house rapid safety evaluation (ATC-45 style) to find unsafe homes and people needing shelter.

Standard procedure

- 1

Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2

Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3

Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4

Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5

Roof work needs fall protection (harness, anchors or edge protection).
- References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-06 Cluster of small structures collapsedP1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:30.0 (seen in 1 frame(s), from 00:30.0)
Component	small buildings and sheds, centre
Extent	Block of ~5 small structures
Category	structural damage (62% confidence)
Severity	Critical, score 3.98 / 4
Safety / environmental hazard	89% probability
Likelihood it is a real defect	93%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Structural engineer + building official; demolition or shoring contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A group of small buildings and sheds in the middle of the block has collapsed or lost its roofs, leaving a jumble of sheets and timber.

Visual evidence: Tangle of roof sheets, timber and broken walls between two larger buildings.

How to fix it

For this finding: Keep people out; clear the collapsed material and check for anyone sheltering there.

Standard procedure

- 1 Cordon off a fall zone around the structure (at least its height) and post it as unsafe.
- 2 Get a structural engineer's rapid safety evaluation (inspected / restricted use / unsafe placard).
- 3 Shore or brace if the structure is to be saved; otherwise plan a controlled demolition.
- 4 Do not let anyone climb or enter until the engineer clears it.

References: ATC-45 (safety evaluation after windstorms and floods); FEMA P-2055 (post-disaster building safety evaluation); IEBC

F-07 Roof blown off, top-floor framing exposed

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:03.0 (seen in 1 frame(s), from 00:03.0)
Component	two-storey house, centre
Extent	1 house, whole roof
Category	roof damage (100% confidence)
Severity	Critical, score 3.94 / 4
Safety / environmental hazard	86% probability
Likelihood it is a real defect	91%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The roof of this house is gone and the top floor's walls and framing are exposed.
Visual evidence: Brown exposed framing and open walls where the roof should be; roofing debris on the slope below.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-08 Roof stripped to exposed rafters

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:30.0 (seen in 1 frame(s), from 00:30.0)
Component	teal two-storey building, lower centre
Extent	1 roof
Category	roof damage (100% confidence)
Severity	Critical, score 3.91 / 4
Safety / environmental hazard	81% probability
Likelihood it is a real defect	94%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

This building's roof covering has been torn off, leaving the rafters exposed to the weather.
Visual evidence: Bare roof framing with no sheeting on a teal-walled building.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-09 Roof stripped to exposed rafters

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:30.0 (seen in 1 frame(s), from 00:30.0)
Component	building at the bottom of the frame
Extent	1 roof
Category	roof damage (100% confidence)
Severity	Critical, score 3.94 / 4
Safety / environmental hazard	85% probability
Likelihood it is a real defect	95%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A second building nearby also has its roof framing exposed.
Visual evidence: Bare rafters with no roof covering.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-10 Building roof gone, interior exposed

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:33.0 (seen in 2 frame(s), from 00:33.0)
Component	two-storey building, top centre
Extent	1 building
Category	roof damage (100% confidence)
Severity	Critical, score 3.88 / 4
Safety / environmental hazard	81% probability
Likelihood it is a real defect	91%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

This building has lost its roof and the upper rooms are open to the sky.
Visual evidence: Orange-brown interior walls visible from above where the roof should be.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-11 Informal structures with damaged roofs

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:33.0 (seen in 1 frame(s), from 00:33.0)
Component	small buildings, right edge
Extent	Several small structures
Category	roof damage (100% confidence)
Severity	Critical, score 3.83 / 4
Safety / environmental hazard	79% probability
Likelihood it is a real defect	86%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Roofing contractor + structural engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A row of small informal buildings on the right has damaged or missing roofing.
Visual evidence: Crumpled and missing roof sheets on low structures.

How to fix it

For this finding: Tarp the opening to stop rain getting in, keep people out of the top floor until an engineer checks the roof framing, then re-roof with hurricane-rated fixings.

Standard procedure

- 1 Keep people out of rooms under the damaged roof until an engineer or building official has checked it.
- 2 Make it weather-tight: cover openings with secured tarps or temporary sheeting (fixed to the structure, not weighed down loosely).
- 3 Have a structural engineer check rafters, trusses and connections under the damaged area before repair.
- 4 Replace damaged decking and roofing; upgrade fixings to current wind-load requirements.
- 5 Roof work needs fall protection (harness, anchors or edge protection).

References: International Existing Building Code (IEBC); ASCE 7 (wind loads); OSHA 29 CFR 1926.501 (fall protection)

F-12 Hurricane debris field across hillside homes

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:00.0 (seen in 2 frame(s), from 00:00.0)
Component	hillside residential area
Extent	Whole hillside in view, hundreds of metres
Category	debris and loose material (100% confidence)
Severity	Critical, score 3.71 / 4
Safety / environmental hazard	80% probability
Likelihood it is a real defect	96%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Roofing sheets, timber and building fragments are scattered across the whole hillside between the homes. Much of this can be picked up by the next strong wind, and it blocks paths and drainage.

Visual evidence: Hundreds of white and grey fragments and broken timber spread over the brown slope and along the road.

How to fix it

For this finding: Clear loose roofing sheets, timber and debris from around occupied homes and roads first; separate hazardous material for proper disposal.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
 - 2 Clear debris from access routes, drains and around buildings.
 - 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.
- References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

F-13 House damaged with debris around the lower floorP1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:33.0 (seen in 1 frame(s), from 00:33.0)
Component	house, centre right
Extent	1 house
Category	debris and loose material (42% confidence)
Severity	Critical, score 3.73 / 4
Safety / environmental hazard	74% probability
Likelihood it is a real defect	89% - VERIFY ON SITE before work
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The lower part of this house is damaged and debris is piled around it.
Visual evidence: Broken lower walls and a heap of debris beside the house.

How to fix it

For this finding: Clear loose roofing sheets, timber and debris from around occupied homes and roads first; separate hazardous material for proper disposal.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
- 2 Clear debris from access routes, drains and around buildings.
- 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.

References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

F-14 Homes destroyed beside the main road

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:15.0 (seen in 1 frame(s), from 00:15.0)
Component	homes along the road, lower right
Extent	2-3 buildings
Category	debris and loose material (83% confidence)
Severity	Critical, score 3.58 / 4
Safety / environmental hazard	53% probability
Likelihood it is a real defect	89%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A group of buildings by the road is largely destroyed, with debris spread onto the verge.
Visual evidence: Broken structures and a scatter of coloured debris next to the road.

How to fix it

For this finding: Clear debris from the road verge and placard the buildings.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
- 2 Clear debris from access routes, drains and around buildings.
- 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.

References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

F-15 Boat stranded on the beach**P1 · Immediate (within 24-72 h)**

Full frame with approximate location



Zoomed detail

Timestamp	00:24.0 (seen in 1 frame(s), from 00:24.0)
Component	Cruz Bay beach
Extent	1 boat
Category	debris and loose material (78% confidence)
Severity	Major, score 3.00 / 4
Safety / environmental hazard	49% probability
Likelihood it is a real defect	87%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A boat has been driven up onto the beach.

Visual evidence: White hull lying on the sand at the water's edge.

How to fix it

For this finding: Check the hull for fuel leaks and have it salvaged.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
- 2 Clear debris from access routes, drains and around buildings.
- 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.

References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

F-16 Public water storage tank needs post-storm inspectionP2 · Planned (within 30 days)



Full frame with approximate location



Zoomed detail

Timestamp	00:36.0 (seen in 1 frame(s), from 00:36.0)
Component	steel water storage tank
Extent	1 tank
Category	utilities damage (96% confidence)
Severity	Major, score 2.77 / 4
Safety / environmental hazard	31% probability
Likelihood it is a real defect	68%
First action	Keep out / make safe: Keep people out and make the area safe now (evacuate, fence off, placard) because of danger to life or of collapse.
Crew	Utility company / licensed electrician; tank inspector
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The large steel water tank that supplies the area has no obvious damage to its shell from the air, but the handrail or equipment on its roof looks bent, and debris lies around its base. Hurricane winds can damage tank roofs and vents where contamination gets in.

Visual evidence: Intact cylindrical shell; a bent-looking rail or frame on the roof edge; scattered debris at the base.

How to fix it

For this finding: Have the utility inspect the tank roof, hatches and vents for openings, and test the stored water before it goes back into supply.

Standard procedure

- 1 Treat all downed or leaning lines as live; keep people away and call the utility.
 - 2 Have the utility or a licensed electrician make safe and restore service.
 - 3 Inspect water tanks for shell, roof and foundation damage before refilling.
- References: NFPA 70 (National Electrical Code); AWWA D100 / D103 (water storage tanks)

F-17 Debris washed up along the beach and waterfrontP2 · Planned (within 30 days)



Full frame with approximate location



Zoomed detail

Timestamp	00:24.0 (seen in 1 frame(s), from 00:24.0)
Component	Cruz Bay beach and seawall
Extent	~100 m of beach
Category	debris and loose material (100% confidence)
Severity	Moderate, score 2.15 / 4
Safety / environmental hazard	36% probability
Likelihood it is a real defect	89%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Debris removal crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Debris has collected along the beach and at the base of the waterfront buildings.

Visual evidence: Pile of timber, fronds and fragments along the top of the beach below the buildings.

How to fix it

For this finding: Clear loose roofing sheets, timber and debris from around occupied homes and roads first; separate hazardous material for proper disposal.

Standard procedure

- 1 Secure or remove loose sheet metal and debris that wind can pick up.
- 2 Clear debris from access routes, drains and around buildings.
- 3 Separate hazardous debris (asbestos-cement sheets, treated wood, chemicals) for proper disposal.

References: FEMA Public Assistance debris guidance; EPA asbestos (NESHAP) rules for demolition debris

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (one every 3.0 s, 13 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Footage: "U.S. Navy conducts damage assessments in the U.S. Virgin Islands", US Navy, 13 Sept 2017, via Wikimedia Commons (CC BY 3.0).